

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT3 — VISUALIZATION CRITIQUE & REDESIGN

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Critique & Redesign
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Evaluate flawed or misleading data visualizations, critique their specific shortcomings, and justify an improved redesign.(BL5)</i>		
Student Name:	Sania Herbert	Reg. No.:	192324062
Section / Batch:	2023/AI&DS	Date:	18.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond simply naming a flaw: explain WHY it is misleading and what a reader could wrongly conclude from it.
- Your redesign must be a specific, justified chart proposal — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION CRITIQUE & REDESIGN

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT3 — Visualization Critique & Redesign

CO3 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO3-AT3 Visualization Critique & Redesign problems, in which students are given a flawed or misleading chart, must critique its specific shortcomings, and propose a justified redesign.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the specific visualization principle being violated (e.g., proportional ink, axis scaling, color perception, overplotting) in the given flawed chart.	Good understanding of the relevant principle, with only minor gaps in identifying exactly what is violated.	Basic understanding, but with some misconceptions about which principle applies to the flaw shown.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	25%	Clearly explains why the flaw is misleading, including what a reader could wrongly conclude as a result — not just that something is 'wrong.'	Explains the flaw with adequate reasoning, covering most of why it misleads a reader.	Identifies the flaw but explanation is generic or only loosely tied to its misleading effect.	Little or no explanation of why the flaw is misleading; the critique is a bare assertion.
Application	20%	Proposes a specific, well-justified redesign that directly resolves the identified flaw, correctly applying	Proposes a workable redesign that resolves the flaw, with only minor gaps in justification.	Proposes a redesign, but it only partially resolves the flaw or lacks sufficient justification.	Fails to propose a workable redesign, or the redesign does not address the identified flaw.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		appropriate chart-selection and design principles.			
Organization	15%	The response is clearly structured, addressing both the critique and the redesign in a logical sequence with coherent overall flow.	The response is organized and addresses both parts, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the critique and the redesign are completely accurate and well-suited to the specific chart and data described.	Mostly accurate, with only minor omissions or a small error in the critique or redesign.	Partially correct; the critique or redesign contains notable inaccuracies or mismatches to the scenario.	Largely incorrect or inappropriate, with major gaps in both the critique and the redesign.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 5

Scenario: A pie chart uses red, orange, and yellow for three nearly-equal-sized categories, making the slices hard to visually tell apart at a glance.

a) Critique this color choice.

Using red, orange, and yellow for nearly equal-sized slices is a poor choice because these colors are very similar in hue and brightness. The slices can be difficult to distinguish quickly, especially for people with color-vision deficiencies. This reduces the chart's readability and makes comparison

difficult.

b) Redesign: propose a better color scheme and explain your reasoning.

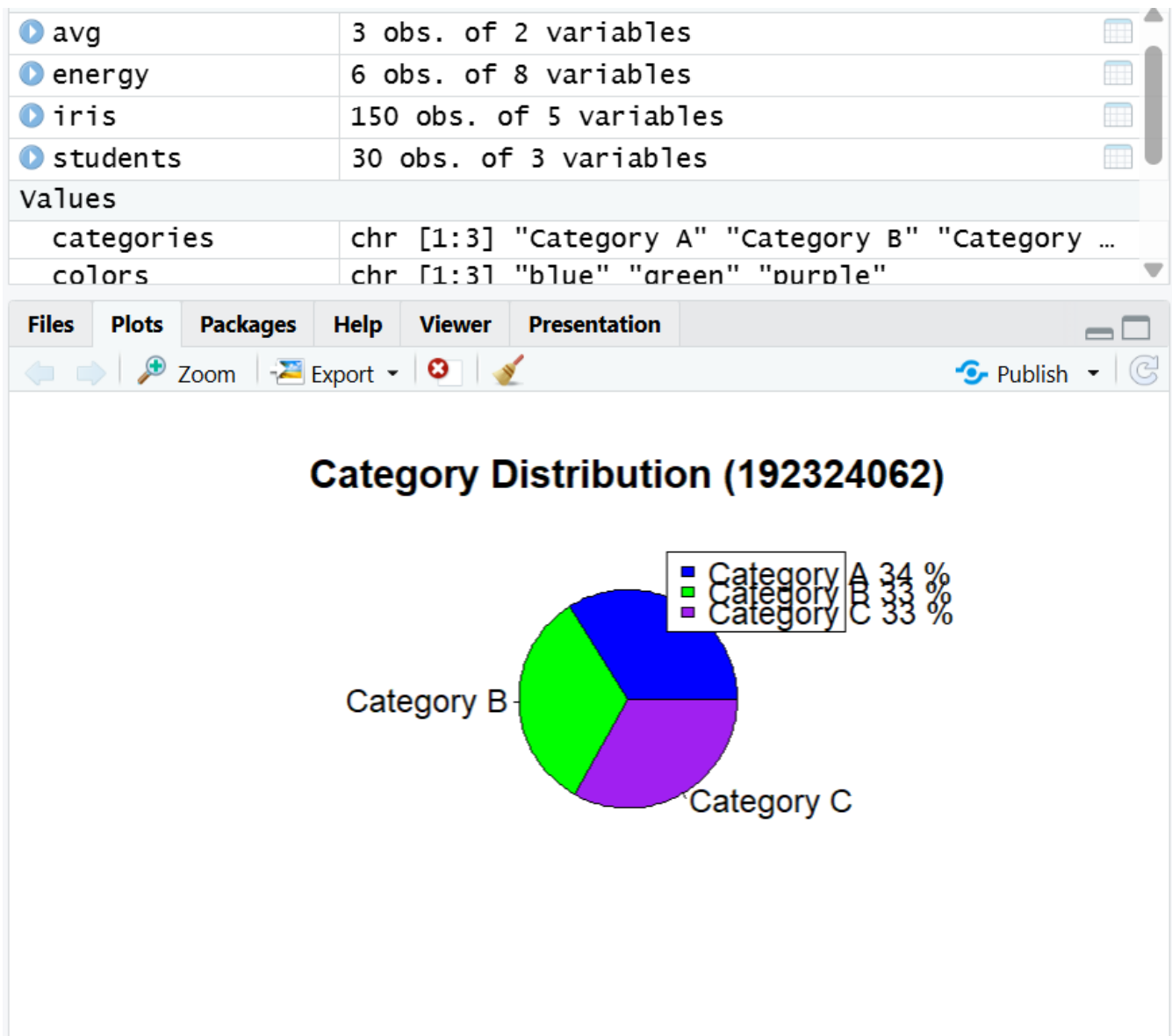
Use clearly contrasting, color-blind-friendly colors such as **blue, green, and purple**. Keep the colors distinct in hue and brightness, and add **data labels or percentages** to the slices. This makes the categories easier to identify and compare accurately.

```
1 # Problem 5: Better Color Scheme
2
3 categories <- c("Category A", "Category B", "Category C")
4 values <- c(34, 33, 33)
5
6 # Better contrasting colors
7 colors <- c("blue", "green", "purple")
8
9 pie(values,
10     labels = categories,
11     col = colors,
12     main = "Category Distribution (192324062)")
13
14 # Add percentages
15 percent <- round(values / sum(values) * 100, 1)
16
17 legend("topright",
```

19:22 (Top Level) R Script

R 4.6.1 · ~/

```
>
> categories <- c("Category A", "Category B", "Category C")
> values <- c(34, 33, 33)
>
> # Better contrasting colors
> colors <- c("blue", "green", "purple")
>
> pie(values,
+     labels = categories,
+     col = colors,
+     main = "Category Distribution (192324062)")
>
> # Add percentages
> percent <- round(values / sum(values) * 100, 1)
```



Problem 6 (5 Marks)

Scenario: A report converts department gender data into a 100% stacked bar chart without showing the total headcount of each department.

a) Critique what critical information is hidden by this design.

A 100% stacked bar chart shows only the percentage distribution of male and female employees in each department. It hides the total headcount, so a department with 10 employees and another with 1,000 employees may appear equally large. This can give a misleading impression of workforce size.

b) Redesign: propose how to present the data so the hidden information becomes visible.

Use a logarithmic x-axis because household income is highly right-skewed. A log scale spreads out smaller income values and compresses the extreme tail, making the underlying distribution easier to see and compare.

```
1 # Problem 6 - 100% Stacked Bar Chart
2
3 departments <- c("HR", "IT", "Finance", "Sales")
4
5 male <- c(30, 80, 40, 70)
6 female <- c(20, 40, 30, 50)
7
8 # Combine data
9 gender_data <- rbind(male, female)
10
11 # Convert to percentages
12 gender_percent <- prop.table(gender_data, margin = 2) * 100
13
14 # 100% stacked bar chart
15 barplot(gender_percent,
16         beside = FALSE,
17         names.arg = departments,
```

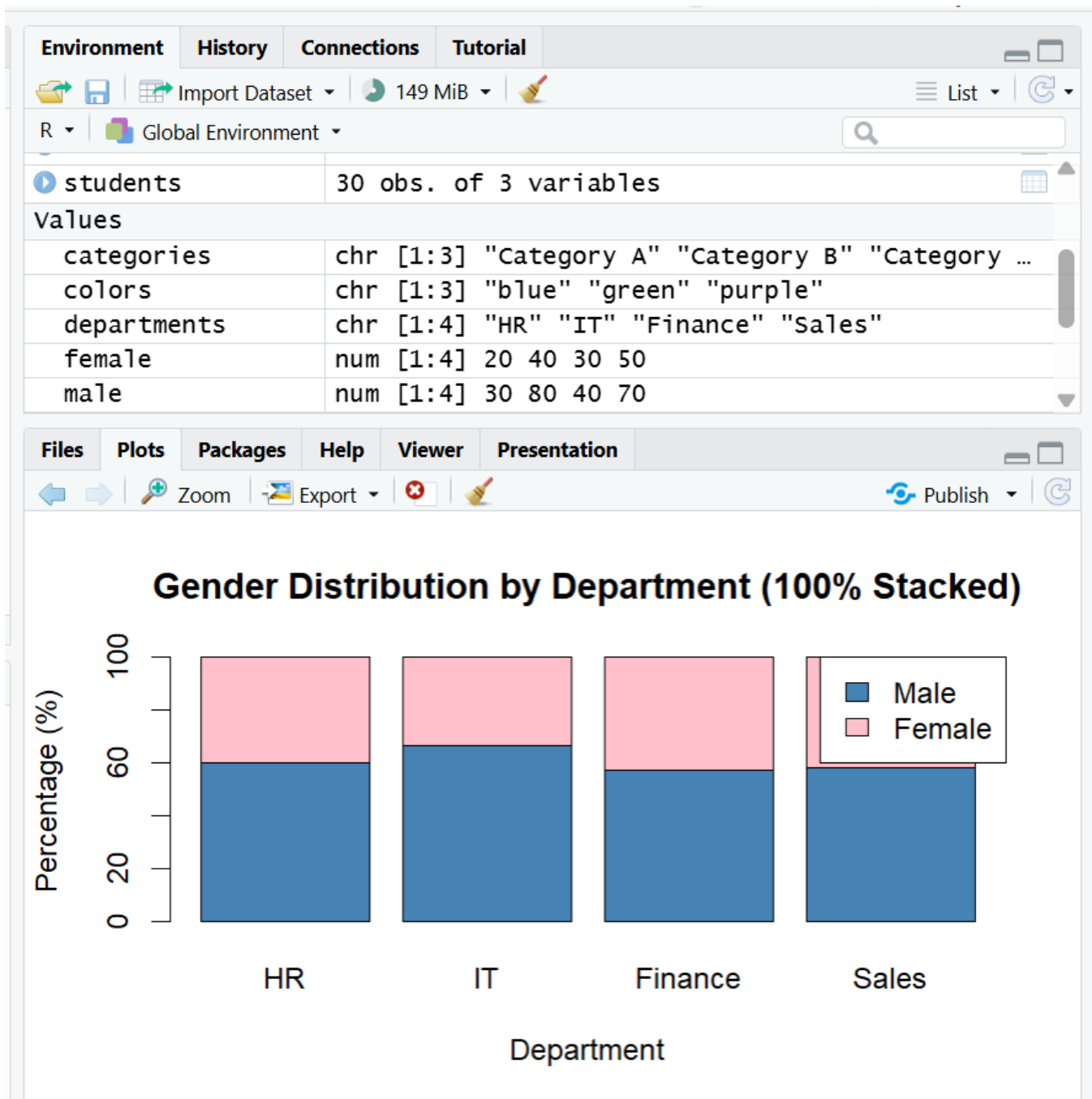
25:38

(Top Level) ↕

R Script

R 4.6.1 · ~/

```
> # 100% stacked bar chart
> barplot(gender_percent,
+         beside = FALSE,
+         names.arg = departments,
+         col = c("steelblue", "pink"),
+         main = "Gender Distribution by Department (100% Stacked)",
+         xlab = "Department",
+         ylab = "Percentage (%)")
>
> legend("topright",
+       legend = c("Male", "Female"),
+       fill = c("steelblue", "pink"))
>
```



Problem 18 (5 Marks)

Scenario: A scatterplot of 100,000 transaction records is shown as a single dense black blob due to heavy overplotting, hiding any underlying pattern.

a) Critique why this chart fails to communicate anything useful.

Plotting 100,000 transactions as individual points causes severe overplotting, producing a dense black blob. Individual observations and underlying relationships or clusters cannot be identified, so the scatterplot fails to

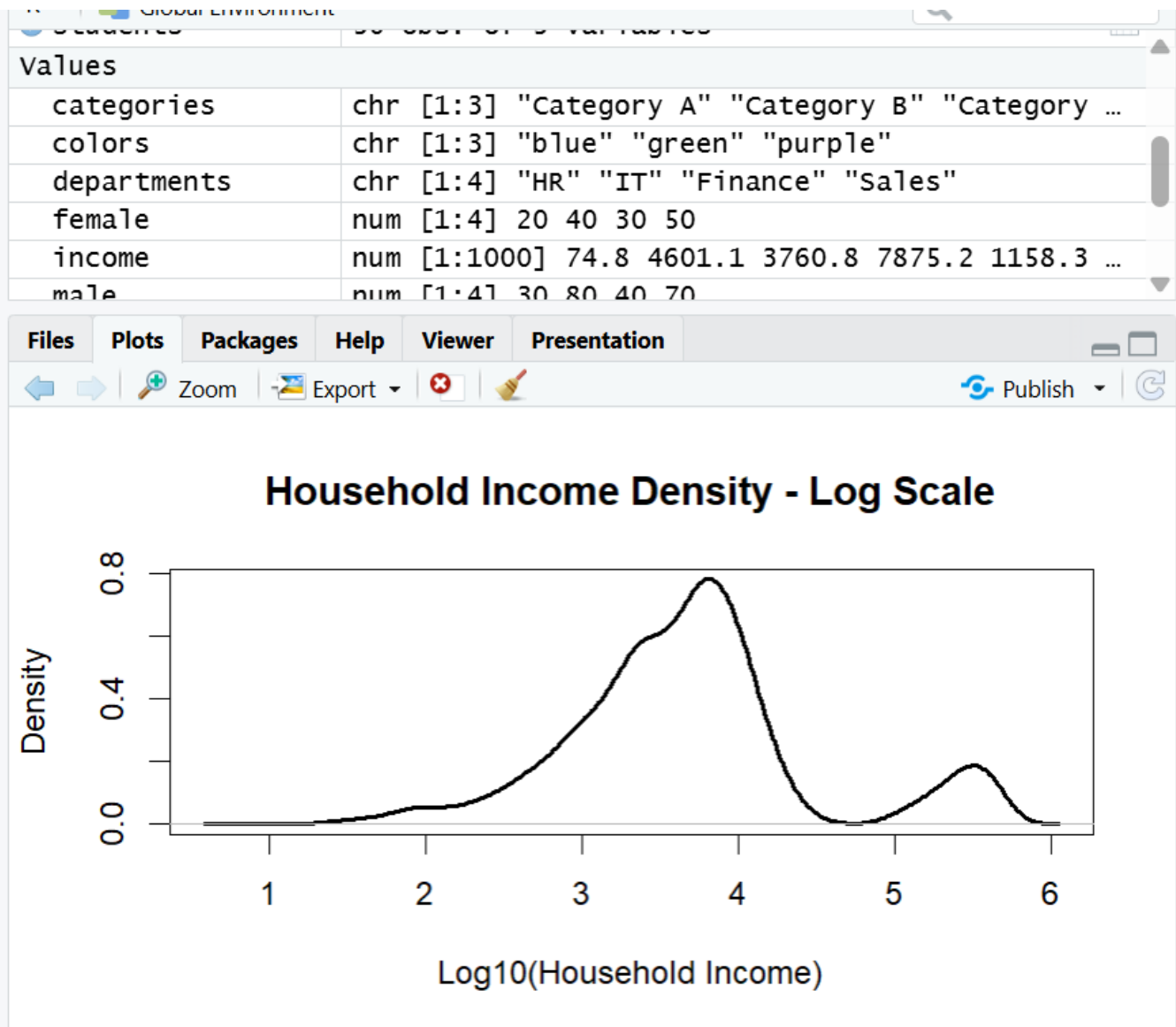
communicate useful information.

b) Redesign: propose an alternative chart or technique that would handle this volume of data.

Use a hexbin plot or 2D density/heatmap to group nearby observations into regions and represent their frequency using shading. This reduces overplotting and makes clusters, concentration areas, and overall patterns visible even with 100,000 records.

```
1 # Problem 12 - Household Income Density Plot
2
3 set.seed(10)
4
5 # Create right-skewed household income data
6 income <- c(
7   rexp(900, rate = 0.0002),
8   runif(100, 100000, 500000)
9 )
10
11 # Density plot
12 plot(density(income),
13       main = "Household Income Density Plot",
14       xlab = "Household Income",
15       ylab = "Density",
16       lwd = 2)
17 # Improved density plot using log scale
```

```
R R 4.6.1 · ~/
> plot(density(income),
+       main = "Household Income Density Plot",
+       xlab = "Household Income",
+       ylab = "Density",
+       lwd = 2)
> # Improved density plot using log scale
>
> plot(density(log10(income)),
+       main = "Household Income Density - Log Scale",
+       xlab = "Log10(Household Income)",
+       ylab = "Density",
+       lwd = 2)
> |
```

Problem 24 (5 Marks)

Scenario: A sunburst chart is used to display 6 hierarchical levels of a company's organizational structure, resulting in tiny, unreadable outer rings.

a) Critique the specific problem with using this many hierarchy levels in a sunburst chart.

A sunburst chart with six hierarchy levels creates very thin outer rings and tiny segments. The labels become unreadable and it becomes difficult to follow relationships between different organizational levels. Therefore, the chart becomes visually cluttered.

b) **Redesign:** propose an alternative approach for visualizing deep hierarchies.

Use a treemap or an indented tree diagram for the deep hierarchy. A tree

diagram is especially suitable because each level can be displayed clearly with indentation and labels. Alternatively, show the organization in multiple simpler charts, one level at a time.

```
1 # =====
2 # PROBLEM 18 - HEXBIN PLOT
3 # 100,000 TRANSACTION RECORDS
4 # =====
5
6 # Load hexbin package
7 library(hexbin)
8
9 # Reproducible results
10 set.seed(10)
11
12 # Generate 100,000 transaction records
13 transaction_amount <- rnorm(
14   100000,
15   mean = 500,
16   sd = 100
17 )
18
19 # Create hexbin data
20 hex_data <- hexbin(
21   transaction_amount,
22   +   customer_spending,
23   +   xbins = 50
24   + )
25 >
26 > # Display hexbin plot
27 > plot(
28   +   hex_data,
29   +   main = "Transaction Pattern Using Hexbin Plot",
30   +   xlab = "Transaction Amount",
31   +   ylab = "Customer Spending"
32   + )
33 >
```

female	num [1:4] 20 40 30 50
income	num [1:1000] 74.8 4601.1 3760.8 7875.2 1158.3 ...
male	num [1:4] 30 80 40 70
mu	5.84333333333333
percent	num [1:3] 34 33 33
sigma	0.828066127977863
transaction_amou...	Large numeric (100000 elements, 800 kB)

